Educational Inequality in U.S. High Schools: The Influence of Socioeconomic and Geographic Factors

Abstract

This project investigates how socioeconomic conditions and geographic location are related to academic performance in U.S. high schools. The analysis included 7,227 high schools across 20 states, using average ACT scores and socioeconomic factors such as unemployment rates, college education levels, household income, marriage rates, and free or reduced-price lunch eligibility. After cleaning the data, I used simple, quadratic, and multiple linear regression to examine these relationships. The socioeconomic model explained 62.8% of the variation in ACT scores, with free or reduced-price lunch eligibility, college education levels, and unemployment rates showing statistically significant relationships. Adding states to the model increased the explained variation to 67.9% and reduced the Mean Absolute Error from 1.15 to 1.03 ACT points. Overall, the findings suggest that both socioeconomic conditions and geographic differences are associated with academic performance, highlighting the importance of considering broader community characteristics when studying educational inequality.

Introduction

Educational opportunities are not the same for every student in the United States. While some students have access to more resources and academic support, others face challenges related to their socioeconomic backgrounds. Understanding how these differences are associated with academic performance is important because it can help identify patterns of educational inequality.

This project examines average ACT scores across U.S. high schools using data from EdGap and the National Center for Education Statistics (NCES). The datasets contain school information and socioeconomic characteristics, including unemployment rates, college education levels, marriage rates, median household income, and free or reduced-price lunch eligibility.

The main goal of this study was to investigate how socioeconomic factors are related to average ACT scores in U.S. high schools. More specifically, I wanted to determine which factors had the strongest relationships with academic performance and whether including states in the analysis improved the regression model. By examining both socioeconomic and geographic differences, this project aims to better understand patterns associated with educational inequality.

Theoretical Background

Educational opportunities are not always equal, as students come from different socioeconomic backgrounds that may influence their academic performance. Factors such as household income, unemployment, college education levels, and family characteristics may be related to the resources and support available to students. Free or reduced-price lunch eligibility can also serve as an indicator of economic disadvantage.

Geographic location is another important consideration. According to the National Center for Education Statistics (NCES, n.d.), public schools receive funding from federal, state, and local sources, with the distribution varying across states. These differences raise questions about how educational resources and opportunities may vary depending on location. By including states in the regression model, this project examines whether geographic information helps explain differences in ACT scores beyond socioeconomic characteristics.

Methodology

The datasets used in this project contained information about U.S. high schools, including average ACT scores and socioeconomic characteristics of their communities. The datasets were combined using school identifiers and filtered to include only high schools. After merging, I checked for duplicates, missing values, and incorrect entries. Invalid negative values were replaced with missing values, and schools without average ACT scores were removed. Missing socioeconomic values were estimated using the Iterative Imputer method in Python. I also used boxplots and the interquartile range (IQR) to identify potential outliers, keeping values that appeared reasonable.

After cleaning the data, I used descriptive statistics, scatterplots, and correlation analysis to explore relationships between socioeconomic factors and ACT scores. I first applied simple linear regression to examine the relationship between median household income and average ACT scores, followed by quadratic regression to check for a possible nonlinear relationship.

Next, I developed a multiple linear regression model using five socioeconomic predictors: unemployment rate, percentage of adults with a college education, marriage rate, median household income, and percentage of students eligible for free or reduced-price lunch. I then added states as categorical variables using dummy encoding, with Illinois as the reference state. I also created a reduced model containing only the three statistically significant socioeconomic predictors and standardized the variables to compare their relative relationships with ACT scores.

Finally, I evaluated the models using R^2 and Mean Absolute Error (MAE). I used an ANOVA test to compare the socioeconomic model with the model including states and examined regression residuals to assess model fit.

Computational Results

The final dataset included 7,227 high schools across 20 U.S. states. Figure 1 shows the geographic distribution of these schools, with the color scale representing the number of schools included from each state.

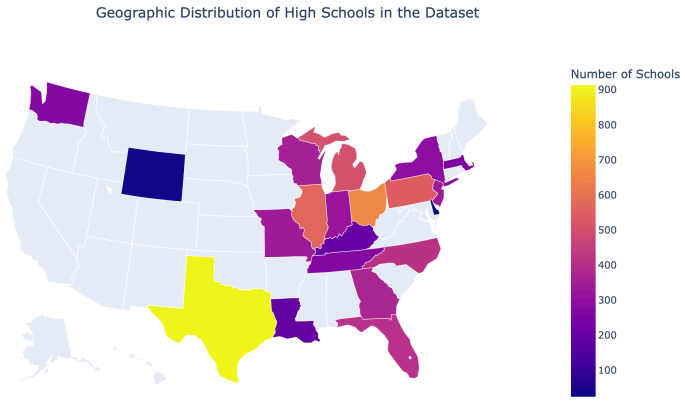


Figure 1. Geographic distribution of high schools included in the dataset, showing the number of schools represented in each state.

A simple linear regression between median household income and average ACT scores produced an R^2 of 0.212. The quadratic regression model produced an R^2 of 0.220. Figure 2 displays the relationship between median household income and average ACT scores, including the fitted linear regression line.

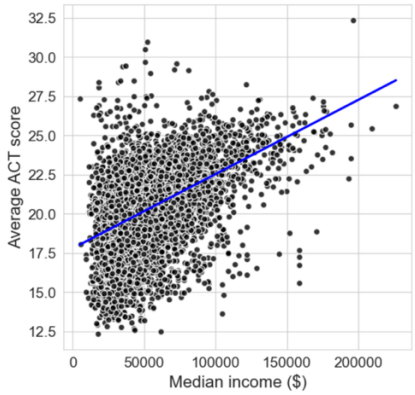


Figure 2. Relationship between median household income and average ACT scores across high schools. Each point represents a school, and the line represents the fitted linear regression.

The multiple linear regression model included unemployment rates, college education levels, marriage rates, median household income, and free or reduced-price lunch eligibility. The model produced an R^2 of 0.628. Table 1 presents the regression coefficients, standard errors, and p-values for each predictor.

	Variable	Coefficient	Std. Error	P-value
0	Intercept	22.6896	0.1375	<0.001
1	rate_unemployment	-2.2842	0.4040	<0.001
2	percent_college	1.7430	0.1575	<0.001
3	percent_married	-0.0752	0.1338	0.574
4	median_income	-4.75e-08	1.21e-06	0.969
5	percent_lunch	-7.5983	0.0968	<0.001
6	R-squared	0.6280		

Table 1: Multiple Linear Regression Results

When comparing average ACT scores across the 20 states, I also noticed some interesting differences. Wisconsin had the highest average ACT score (22.02), followed by Massachusetts (21.86) and Washington (21.39). On the other hand, Delaware had the lowest average (17.39), followed by New York (17.66) and Louisiana (18.91). These averages reflect the schools included in the dataset rather than all schools within each state.

The analysis was extended by including states as categorical variables. A reduced socioeconomic model was also fitted using unemployment rates, college education levels, and free or reduced-price lunch eligibility. Table 2 compares the R^2 and Mean Absolute Error (MAE) of the three multiple regression models.

	Mean Absolute Error	R-squared
Socioeconomic Model Normalized	1.1453	0.6280
Reduced model	1.1453	0.6280
Socioeconomic + States normalized	1.0276	0.6792

Table 2: Regression Model Performance Comparison

An ANOVA comparison between the socioeconomic model and the model including states produced a statistically significant result ($p < 0.001$). The MAE and R^2 values reported above were calculated using the data used to fit the models.

Discussion

The results suggest that socioeconomic conditions are strongly related to ACT performance in U.S. high schools. Free or reduced-price lunch eligibility had the strongest relationship with ACT scores, followed by college education levels and unemployment rates. One interesting finding was that median household income showed a positive relationship with ACT scores in the simple linear regression, but was no longer statistically significant when other factors were included. This suggests that income alone may not be as informative when other characteristics of the community are considered. The reduced model also performed almost identically to the full socioeconomic model, showing that removing variables that were not statistically significant did not noticeably affect its performance.

One finding that stood out to me was how including states improved the regression model, increasing R^2 from 0.628 to 0.679 and reducing MAE from approximately 1.15 to 1.03 ACT points. I also learned that public education funding comes from different levels of government and that the distribution varies across states (NCES, n.d.). This made me wonder whether differences in school funding could help explain some of the geographic variation. However, since funding was not directly analyzed, I cannot determine whether it influenced ACT scores.

There are also some limitations to consider. The dataset only included 20 states, with different numbers of schools represented in each, which limits how broadly the findings can be applied. ACT participation may also vary between states, potentially affecting the comparisons. Additionally, some missing socioeconomic values were estimated during data cleaning, which may introduce uncertainty. Although including states reduced the model's error, an MAE of approximately 1.03 means the fitted ACT scores still differed from the observed scores by about one point on average. Finally, these relationships do not establish causation, meaning the results cannot prove that socioeconomic conditions or geographic location directly determine academic performance.

Conclusion

This project explored how socioeconomic conditions and geographic location are associated with differences in ACT performance across U.S. high schools. The results showed that factors such as free or reduced-price lunch eligibility, college education levels, and unemployment rates were significantly related to ACT scores. Including states in the regression model also improved its performance, suggesting that location provides additional information beyond socioeconomic conditions alone. Although the analysis was limited to 20 states, these findings highlight how educational outcomes may be connected to the communities and environments where students live. Understanding these relationships is important because it can help identify patterns of educational inequality and guide future research into the challenges students face. Ultimately, improving educational opportunities requires looking beyond individual student performance and considering the broader conditions that may shape their education.

References

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